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The spatiotemporal trends of urban-rural green spaces and their heterogeneous relationships with population and economic vitality: Evidence from the Red River Delta, Vietnam

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Abstract:

The preservation of green spaces in rapid urbanization emerges as a critical concern, posing a significant challenge in fostering green cities. This research centres on the Red River Delta, employing it as a case study to delve into the intricate interplay between population dynamics, economic vitality, and the maintenance of green spaces, elucidating nuanced effects discernible in urban and rural areas. This study utilizes the Normalized Difference Vegetation Index (NDVI) to depict temporal changes in green space from 2009 to 2022. The results revealed a discernible decline in urban and rural areas in the first eight years and increased slightly afterwards. The results underscore the salient impact of economic and population factors on green space in rural locales, contrasting with the predominant role of population characteristics in urban settings. Specifically, immigration rates and population density emerge as pivotal determinants influencing green spaces in both contexts. The study discloses a heterogeneous association between green spaces and economic development, marked by a positive correlation in urban areas and a negative impact in rural expanses. Moreover, the implications of population and economic vitality on green spaces exhibit variability contingent upon the cities' function. This research extends the scope of urban

development theory. It contributes to the discourse on sustainable urban green development, offering empirical evidence derived from the distinctive context of the Red River Delta in Vietnam.

Keywords: Heterogeneous effects, green space, NDVI, population, economic vitality, Red River Delta, sustainable urban development.

1. Introduction

Green spaces (GS) are crucial in urban development [1, 2]. Firstly, GS exhibit high efficacy in diminishing air pollution [3], addressing the impacts of climate change [4, 5], alleviating the urban heat island effect [6], and fostering biodiversity [7]. Secondly, GS has a positive impact on human health [8], enhancing public health and overall well-being [9]. Additionally, GS provides cultural, aesthetic, and educational services, promotes street vitality through recreational spaces [10], facilitates social interactions [11], and enhances urban landscapes [12]. Consequently, the EU and the US have recommended that local governments provide GS within a 10-minute walk for citizens [13]. Nevertheless, despite the manifold benefits of GS for the environment, society, and individuals, many countries struggle to find a harmonious balance between economic interests and GS development.

On the one hand, developed countries have made efforts to maintain green spaces. For example, between 1990 and 2015, European countries expanded urban centres into higher GS areas, significantly increasing green space per capita [14]. In addition, policies were implemented to protect and develop trees, such as planting new trees and protecting old ones. These policies helped to increase green space, create better living environments, and mitigate the effects of climate change.

On the other hand, cities in developing countries are experiencing a rapid decline in green spaces due to urbanization [15]. Beijing, for instance, is facing environmental problems due to the lack of green spaces and the pressure of economic and population development. Similarly, many Chinese cities also prioritized large-scale housing projects, neglecting the

importance of green spaces [16]. Furthermore, previous studies have shown a significant trend of browning, which refers to converting forest land, agricultural land, and green land into impervious surfaces such as roads, houses, and industries, resulting in increasingly narrow areas for green space [17-19]. Specifically, in the suburbs of Shanghai, the advanced vegetation index and gross primary production (GPP) have significantly decreased from 2000 to 2016 due to urban expansion and economic development [18].

Numerous studies have investigated the association between socio-economic factors and urban green space from global and local perspectives [20]. However, there are still several research gaps that require further exploration. Firstly, existing research primarily focuses on the impact of population density, spending level, and GDP on urban green space [17, 21]. However, other critical factors that contribute to population and economic growth, such as population growth rate, immigration and emigration rates, labour force, foreign investment capital, firms, and income, have received comparatively less attention. Secondly, while significant attention has been given to the effects of population and economy on urban green areas at the provincial level, little is known about the differences in these effects between urban and rural areas. Given the distinct characteristics and dynamics of rural and urban areas, exploring how population and economic factors influence green spaces in these different contexts is essential. Exploring these research gaps can yield valuable insights into the complex relationship between socio-economic factors and green spaces in both urban and rural areas.

Over the past 10 years, green space areas in Vietnam have gradually decreased due to the increasing demand for urban and economic development. In Hanoi, construction projects converted much agricultural land into residential and commercial land [22], leading to adverse effects, such as waterlogging and summer heat shock [23]. For example, the conversion of more than 160 hectares of vegetative land to impervious surface in Hanoi's centre has increased the temperature by nearly 3.3°C, contributing to the hottest heat waves recorded in the Red

River Delta [24]. Despite the adverse effects of the lack of green space on human life, very few studies investigated the relationship between green spaces and population and economic development in Vietnam, as well as developing countries in the first stage of urbanization. Existing studies mainly focus on green space and urban heat island effects [10], urban greening and ecosystem services [25], and green space planning [26]. Hence, it is essential to conduct further research to comprehensively elucidate the influence of population and economic vitality on green space. Utilizing the Red River Delta, Vietnam, as the research area, this study aims to investigate the effects of population and economic vitality on green spaces, focusing on different impacts between urban and rural areas. The specific objectives of this study are to address the following research questions:

(1) How has the spatiotemporal distribution of green space evolved from 2009 to 2022?

(2) What are the relationships between green spaces, population, and economic vitality?

(3) How does the influence of population and economic vitality on green spaces vary between urban and rural areas?

In this study, "green space" refers to various types of vegetated areas, including parks, woods, gardens, farms, and vegetation along transportation corridors [27], which benefits the environment, society, and public health. The term "population and economic vitality" describes the metabolism and growth of a city [28]. Investigating the relationships between green spaces and population and economic vitality, and differences between urban and rural areas will help managers and policymakers better understand the effects of regional economic development on green spaces. It supports policymakers to make informed decisions, balancing green space expansion and economic development. As a result, policymakers can develop appropriate strategies to balance the benefits of population development, economic vitality, and green space conservation.

This article is structured as follows: Section 2 presents the research area, data, and methods. Section 3 presents the main results on the spatiotemporal distribution of green space and the relationships between population, economic vitality, and green space (urban and rural areas). Section 4 discusses the findings, and Section 5 concludes.

2. Study area, data, and methods

2.1. Study area

The study employed the Red River Delta, located in northern Vietnam, as the study area. While this region covers only 6.5% of the country's land area, the population is approximately 23.4 million (23.57% of Vietnam's total population). The Red River Delta has the highest average population density in Vietnam, with 1102 persons per km². With an urban population of 8.8 million, this area constitutes 23.6% of Vietnam's urban population [29]. The region comprises two special cities under central government administration¹ (Ha Noi and Hai Phong), and nine provinces consisting of 16 cities, 19 urban districts, 6 towns, and 88 rural districts. Over the past decade, the region has experienced significant economic growth, with a GDP averaging 7.94% between 2009 and 2020, higher than the national average. In 2022, the Red River Delta has also accumulated foreign direct investment projects totalling 11281 million USD, which accounts for 38.51% of the total foreign direct investment in Vietnam [29]. Therefore, the Red River Delta presents a unique opportunity to investigate the relationship between population, economic vitality, and green space in urban and rural areas.

¹ A type of administrative unit at the province level in Vietnam. These are cities under the management of the central government.

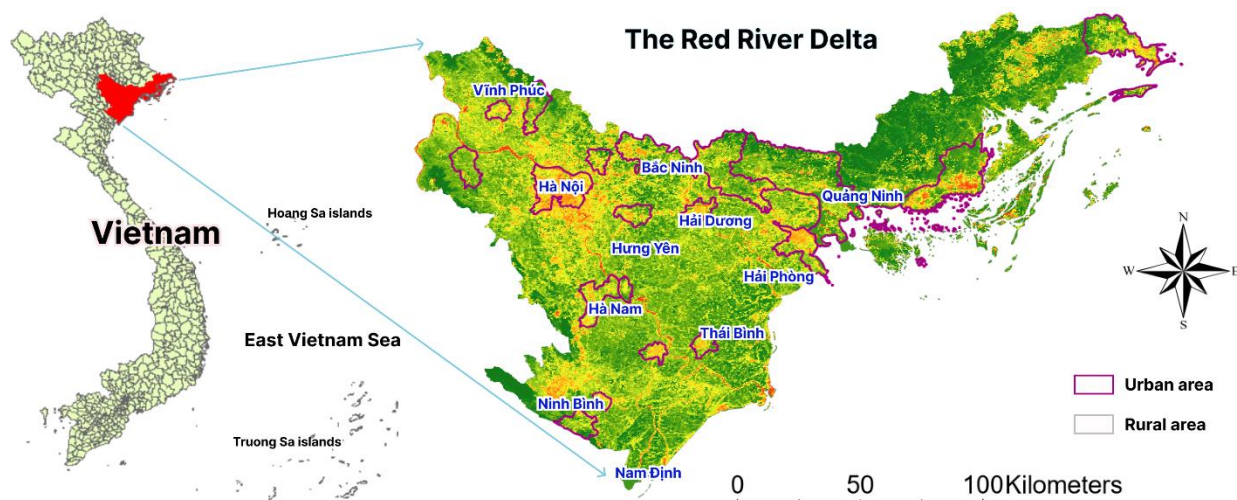


Figure 1. Location of the Red River Delta, Vietnam

2.2. Data

2.2.1. Remote sensing images

Remote sensing data has emerged as an effective means of monitoring land surface changes [30]. High-resolution satellite images with good spectral and spatial resolution were proven effective in accurately monitoring green spaces [31]. Furthermore, medium-resolution multispectral satellite images, including those from Landsat, MODIS, and Sentinel-2, have been employed for monitoring green spaces and computing vegetation indices [32]. Indices derived from satellite data, such as the NDVI and leaf area index, are widely used to detect changes in greenness [32, 33]. Therefore, this study used MODIS MOD13Q1 images from the United States Geological Survey (USGS) website (<https://lpdaac.usgs.gov>) to calculate NDVI values. The raw image data is 16-days data. Before calculating the NDVI values for 2009–2022, the study performed data processing tasks (mosaic, projection, clipping, and maximum synthesis).

The main objective of this study is to examine the impacts of population and economic vitality on urban and suburban green areas. Therefore, the urban-rural boundary was determined based on the administrative unit classification to distinguish the urban-rural areas. Urban areas include the city centers, urban districts, and towns, while the rural areas include

the remaining rural districts. Then, the vegetation difference index was calculated for each study unit.

2.2.2. Weather, population, economic data

Green spaces (GS) are affected by a variety of factors, both natural and humanrelated. Factors such as temperature, rainfall, and air quality significantly impact the health and growth of greenery [34]. Therefore, the study collected data on average temperature and rainfall in the provinces of the Red River Delta from 2009 to 2022 from the World Bank Group (<https://climateknowledgeportal.worldbank.org/>).

This study employs indicators related to population vitality, including population density and growth, as they have been demonstrated to be associated with urban vitality in prior research [21, 35, 36]. Additionally, the study incorporates variables reflecting natural population growth, immigration, and out-migration rates in large urbanized cities [28]. From an economic perspective, new businesses are a crucial driver of economic vibrancy [37]. New businesses contribute to economic growth by providing employment opportunities, which is particularly significant for developing economies. Moreover, foreign investment demonstrates the competitiveness and dynamism of cities, which generates income for businesses and workers and often involves new projects. When cities attract foreign investment, domestic businesses have to compete with foreign ones, which enhances the economy's competitiveness and motivates domestic businesses to improve the quality of their goods and services. Thus, the foreign investment index is widely used in examining economic vitality studies [38]. While Gross Domestic Product (GDP) is a pivotal metric for gauging regional economic development [39, 40], it is not entirely recorded at the Vietnamese provincial level. Consequently, this study employed the volume of goods as a surrogate measure for GDP. This metric is significant as it reflects the value of goods sold in the economy. A discernible augmentation in the aggregate retail sales volume of goods signifies economic expansion and an enhancement in consumers'

purchasing power, thereby contributing to the GDP of counties [38]. Furthermore, the inclusion of the number of non-agricultural business establishments in the regression model is motivated by its substantive role in a nation's GDP dynamics [41]. Non-agricultural businesses create employment opportunities in various sectors, including services, industry, and commerce, and require a substantial workforce to operate effectively. This contributes to the overall economic development of a country by generating employment opportunities and contributing to GDP. Consequently, this metric emerges as a vital component delineating the economic vitality of urban centers [42]. The descriptive statistics of the variables are shown in Table 1 and Figure 2.

Table 1. Descriptive statistics of variables

No	Variables	Definition	Obs	Mean	St.D	Min	Max
Dependent variable							
D1	Urban green index (NDVI) in the whole province	Normalized Difference Vegetation Index in the whole province	154	0.6195	0.0643	0.4773	0.7593
D2	NDVI in urban area	Normalized Difference Vegetation Index in urban area	154	0.5347	0.090	0.2854	0.6920
D3	NDVI in a rural area	Normalized Difference Vegetation Index in rural area	154	0.6372	0.066	0.4767	0.7966
Independent variable							
I1	Population density (persons/km ²)	Number of people per square kilometer	154	1116	482.57	188	2480
I2	The natural growth rate of population (‰)	The difference between live births and deaths relative to the average population	154	9.287	3.0917	1.9	19.2
I3	In-migration rate (‰)	The average number of immigrants per 1,000 residents from several territorial units to a territorial unit	154	5.376	6.5016	0.4	45.7
I4	Out migration rate (‰)	Number out migrants of a territorial unit on average 1000 population	154	5.133	2.6911	0.8	16
I5	Number of enterprises	The number of enterprises were established in a year	154	14224	32042.53	1102	165875
I6	Foreign investment	Amounts of money and kinds brought into Vietnam by foreign investors in the	154	829.1	1273.856	3.5	8669.7

No	Variables	Definition	Obs	Mean	St.D	Min	Max
	capital (Million USD)	form of money, machinery, equipment, etc.					
17	Gross retail sales of goods and services (billion dongs)	The total revenue from selling consumer products and services produced by and provided by businesses.	154	65321	97567.38	1965	512331
18	Non-farm individual business	Executing production, business operations, and service delivery in all economic activities (apart from those in agriculture, forestry, and fisheries)	154	112168	78820.17	47196	385223
19	Workers' income (thousand dongs)	The total amount of income received by employees in a year	154	32759.49	59267.4	1026	301801
110	Temperature (°C)	The average temperature in a year	154	24.78	0.63	22.31	25.90
111	Precipitation (mm)	The average precipitation in a year	154	1602	169.12	2125	1231

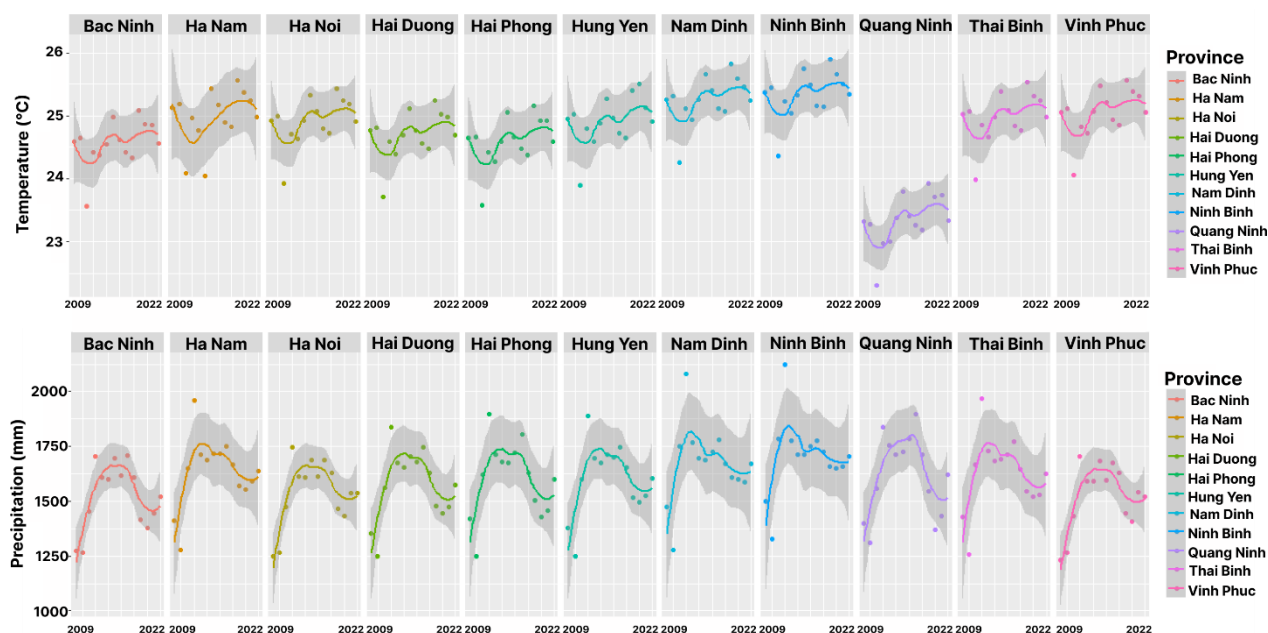


Figure 2. Statistical precipitation and temperature of provinces in the Red River Delta

2.3. Method

2.3.1. The urban green indicator

This study used the 250x250m resolution Normalized Difference Vegetation Index (NDVI) to assess changes in green space. NDVI values range from -1 to 1 and indicate vegetation abundance and health, with higher values indicating healthier vegetation. The NDVI is highly sensitive to vegetation presence and health, making it an effective tool for monitoring

changes in vegetation cover and productivity over time. NDVI is independent of atmospheric effects like clouds and haze, making it ideal for monitoring vegetation in regions with variable weather conditions. Thus, NDVI can detect changes in green space structure and urban greenness. The NDVI (250m x 250m resolution) was widely applied in numerous urban studies [43, 44] and has been proven suitable for residential areas. This paper computed the NDVI value for each city at different times using the MODIS images and determined the average NDVI value of the area as follows:

$$NDVI_{i,t} = \sum_{x=1}^n \frac{NDVI_{x,i,t}}{n_{i,t}} \quad (1)$$

Where, $NDVI_{i,t}$ is the average NDVI of city i in the year t , $NDVI_{x,i,t}$ is the NDVI at x cell of i city, in year t ; $n_{i,t}$ is the total of cells of city i^{th} at year t^{th} . This study extracted the NDVI values for 11 provinces in the Red River Delta from 2009 to 2022, where i ranges from 1 to 11 and j ranges from 2009 to 2022. The analysis began in 2009, marking the first year after the merger of Ha Tay province and some districts of Vinh Phuc to the Hanoi capital. The statistical yearbook has also recorded detailed socio-economic and population data since 2009.

2.3.2. Hedonic model

This study employed stepwise regression analysis to investigate the impact of population and economic vitality on green areas. Stepwise regression is widely used to examine the relationship between independent and dependent variables [45-48]. The model formulation is presented below:

$$G_{it} = \beta_0 + \beta_1 X_{it} + \delta_i + \gamma_t + \varepsilon_{it} \quad (2)$$

Where: G is the dependent variable; $X_{i,t}$ is a matrix of independent variables; β_0 is intercept; β_1 is the coefficient of independent variables; δ_i and γ_t are the fixed effect for types of city and time factors, ε_{it} refers to the random error term. In this study, the matrix $X_{i,t}$ includes population vitality and economic vitality variables.

3. Results

3.1. Spatiotemporal trends of urban green space

The study analyzed the Normalized Difference Vegetation Index (NDVI) to assess the greenery status in the Red River Delta. The average NDVI was calculated for the entire city and urban-rural areas. The findings highlighted a significant difference in urban greenness (UG) among different types of cities. Specifically, the centrally controlled cities (i.e., Ha Noi and Hai Phong) had the lowest average NDVI over the 14 years, with values of 0.5634 and 0.5673, respectively. On the other hand, provinces near Ha Noi, such as Vinh Phuc, Bac Ninh, and Hung Yen, had slightly higher NDVI values ranging from 0.588 to 0.605. The provinces with the highest NDVI values were Quang Ninh, Nam Dinh, Thai Binh, and Ninh Binh. While Quang Ninh and Ninh Binh have mountainous regions, Nam Dinh and Thai Binh are significant rice production centers in Northern Vietnam. This highlights how different provinces influenced the GS. This study used an ANOVA test to compare the average NDVI of provinces over 14 years (Table 2). The results revealed a significant difference in the average NDVI among the 11 provinces studied. Overall, the analysis underscores the need for a comprehensive understanding of the factors affecting GS in urban areas, including natural and human-related factors.

Table 2. The ANOVA test for NDVI of 11 provinces

	Df	Sum sq	Mean Sq	F value	P-value
Province	10	0.2421	0.0242	8.834	3.36e-11 ***
Residuals	143	0.3919	0.0027		
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

This study also examines the NDVI values in both urban and rural areas. The results indicated significant variations in the NDVI values among the urban areas. The Hanoi center records the lowest NDVI value of 0.372, followed by Hai Phong at 0.441. Although the average NDVI values in Nam Dinh and Ha Nam provinces are high, the NDVI values are relatively low in their city center. This phenomenon suggests that the city center of these provinces is

characterized by high-density built-up surfaces such as buildings, pavements, and roads, which have replaced green spaces, as previously reported by Du et al [49]. The lack of green spaces in the city centers is the main cause of the low NDVI values observed. Furthermore, the urban heat island phenomenon contributes to the high temperatures in these areas, which can further reduce vegetation cover and lower the NDVI values.

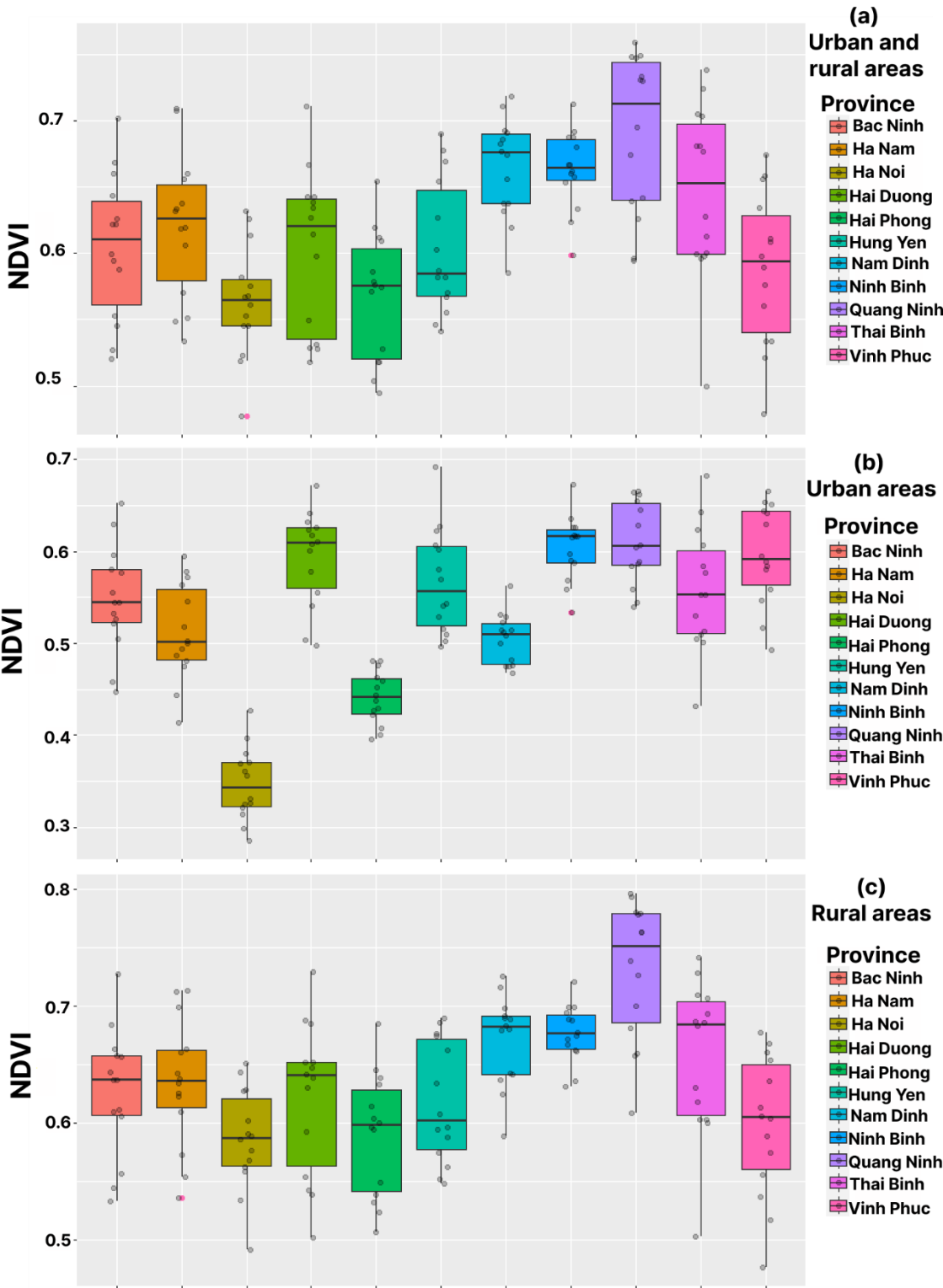
Conversely, the NDVI values in rural areas of most provinces range from 0.58 to 0.68, indicating a relatively higher and consistent vegetation cover. Notably, Quang Ninh stands out with an exceptionally high NDVI value of 0.75, suggesting a rich vegetation cover in this province. An ANOVA test was conducted to determine if there was a significant difference in the average NDVI values of rural areas over the 14 years (Table 3). The results supported the hypothesis that there is a significant difference in the NDVI between rural areas. These findings highlight the disparities in greenness between urban and rural areas in the Red River Delta and emphasize the importance of preserving green spaces in urban centers to mitigate the adverse effects of urbanization and enhance environmental quality.

Table 3. The ANOVA test for rural green area of 11 provinces

	Df	Sum sq	Mean Sq	F value	P-value
Rural area of provinces	10	0.2651	0.0265	9.324	8.53e-12***
Residuals	143	0.4066	0.0028		
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Figure 3 illustrates the average NDVI of provinces across urban, rural, and whole-province scales, providing insights into the vegetation dynamics over the study period. The overall trend of the NDVI showed a declining pattern across all provinces over the study period. The NDVI dropped from 2009 to 2016 before recovering afterwards. Notably, Nam Dinh province demonstrated a decrease in NDVI values in both urban and rural areas over the entire study period. The lowest NDVI value was recorded in the Hanoi center in 2013, with a value of 0.2854. This decline was attributed to a large-scale replacement of 6,700 trees in 190 streets

247 with other urban trees. However, the NDVI values tended to recover in most provinces across
248 urban and rural areas after 2016 (Figure 4).



249 *Figure 3. The average NDVI of provinces divided by urban and rural area*
250 *(from 2009 to 2022)*
251



Figure 4. The changing trend of NDVI from 2009 to 2022, divided by urban and rural areas

Figure 5 shows the spatiotemporal trends of NDVI in the Red River Delta provinces from 2009 to 2022. The color scale, ranging from yellow to dark green, visually represents the increasing greenness over time. There were significant differences in NDVI between provinces, with the lowest and highest values recorded in 2013 and 2011, respectively. From 2009 to 2016, the overall greenness in the region declined, with southeastern provinces experiencing a rapid decline. Among these provinces, Nam Dinh and Ninh Binh had the slowest rate of NDVI decline during this period. Subsequently, from 2018 to 2020, the NDVI increased in most provinces, except for Nam Dinh, which remained stable before slightly decreasing at the end of the period. Noticeably, the northeastern area (i.e., Quang Ninh) experienced fluctuation in greenness in the studied period. The observed variations are attributed to natural events such as floods, wildfires, and construction activities. Extreme weather conditions or natural disasters might lead to temporary reductions in greenness [50], followed by recovery as vegetation

regrows. For example, the historical flood within 55 years (estimated 1500mm in a week) in 2015 was the loss of vegetation in rural areas. Plants that cannot survive in waterlogged conditions may die, especially when the roots are submerged for extended periods, which can cause a significant reduction in NDVI.

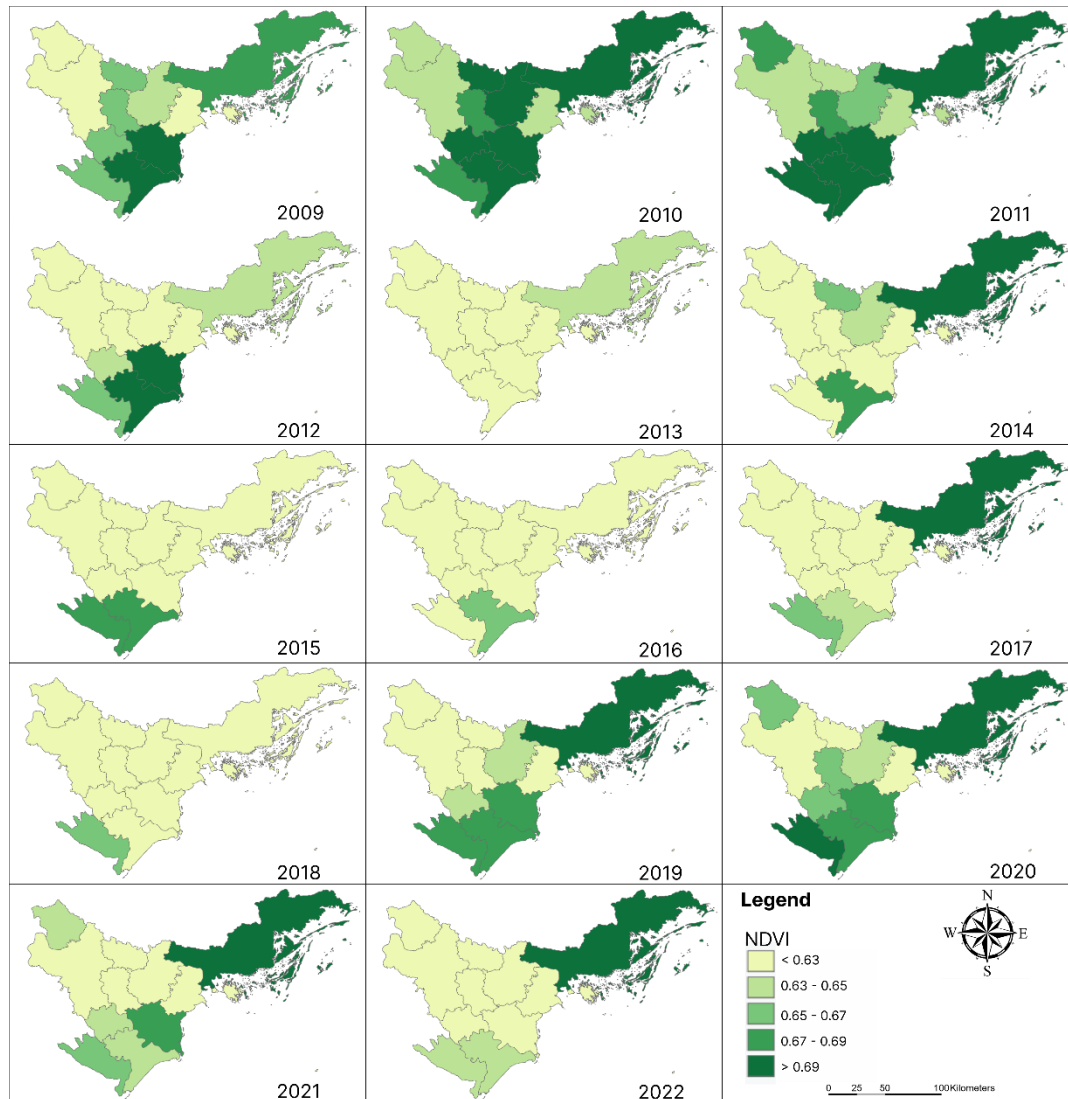


Figure 5. The spatiotemporal distribution of NDVI in the Red River Delta, Vietnam

3.2. The effects of population and economic vitality on greenness

The study examines the relationships between population, economic vitality, and greenness, focusing on the differences between urban and rural areas. Firstly, this paper calculated the correlation between dependent variables (i.e., NDVI of the whole province, NDVI in urban areas, and NDVI in rural areas) and independent variables (i.e., demographic components such as population density, natural population growth, in-migration rate, and out-

migration rate; and economic such as the number of enterprises, foreign investment capital, gross retail sale, non-farm individual business, and workers' income). As indicated by prior studies, temperature and precipitation were employed as control variables due to their established influence on NDVI. To further test these associations, regression models were employed for the whole province (Models 1, 2, and 3), urban areas (Models 4, 5, and 6), and rural areas (Models 7, 8, and 9).

This study used a stepwise approach to explore the influence of population and economic vitality on greenness (Table 4). Model 1 shows a negative relationship between greenness and population density. The rapid population increase intensifies land use demand, impacting green space availability due to various factors such as planning constraints and policies [51, 52]. The escalation in population density also increased vehicles and industrial activity, thus releasing more dust, air pollution, and black smoke [53]. This can reduce the ability of plants to absorb light and CO₂, leading to a decrease in NDVI [54, 55]. Specifically, a 1‰ increase in population density is associated with an approximate 0.049 decrease in greenness.

Furthermore, the out-migration rate significantly contributes to the greenness level. A high out-migration rate indicates migration and population displacement within a given area, decreasing population density, and reducing human demand for green spaces. Additionally, an escalating out-migration rate is associated with agricultural land abandonment and the subsequent regrowth of native vegetation [56, 57]. However, contrasting findings by Gray and Bilborrow [58] indicate a positive relationship between the out-migration rate and green spaces because of offset by migrant remittances. The outcomes reveal that a 1‰ increase in the out-migration rate corresponds to a 0.035 increase in greenness. As population pressure diminishes, the decrease in land use demand creates conditions conducive to an augmented greenness value [59]. The results underscore the statistically significant influence of population

and the out-migration rate on greenness, evident at 99% and 95% confidence levels, respectively. On the other hand, other variables, such as the natural growth rate of the population and the in-migration rate, exhibit negative associations with greenness but do not attain statistical significance.

Model 2 focuses on the impact of economic vitality on greenness, examining key variables such as workers' income, the number of enterprises, and foreign investment capital. The results demonstrate a positive association between workers' income and greenness (coef = 0.0406, p-value < 0.001). Additionally, the number of enterprises has a positive impact on greenness. In contrast, foreign investment negatively affects greenness (coef = -0.0127, p-value < 0.01). The reason is that foreign investment activities often focus on the industrial and construction sectors, resulting in resource exploitation, environmental pollution [60], and a subsequent decline in greenness [49]. Furthermore, foreign investment activities, including land use transition, can lead to landscape changes and resource distribution imbalances, causing significant changes in the local ecosystem [61, 62]. The analysis also reveals variations in the influence of economic vitality across different types of provinces. Centrally administered cities such as Ha Noi and Hai Phong, characterized by superior economic vitality, exert a more pronounced impact on greenness than other provinces.

Model 3 investigated the impact of population and economic vitality on greenness. The findings indicate a direct correlation between population density and out-migration rate with greenness. The study identified a negative association between workers' income and greenness when concurrently assessing the impact of population and economic aspects. As observed in this study, the negative impact of workers' income on greenness can be attributed to complex interactions influenced by urbanization dynamics and changing land-use patterns. In rapidly developing rural areas, a worker's income surge often coincides with increased demands for housing and infrastructure [22]. This surge may result in converting green spaces into

commercial or residential zones to accommodate the growing population and support expanding economic activities [63]. Higher income levels can also elevate property values, prompting developers to prioritize real estate projects over preserving green areas. This observation suggests a discernible discrepancy in the influence of average income between urban and rural areas, consequently resulting in a disparity in the greenness of these regions. Motivated by these findings, the paper further explored the effects of population and economic vitality on greenness in urban and rural areas.

The study employed a similar model to analyze the impact of population and economics on greenness, specifically within urban areas (models 4 to 6). Model 4 highlighted that population density and out-migration rates negatively influence greenness in urban areas. Inadequate housing, infrastructure, limited space, and limited economic opportunities are significant factors that compel individuals to relocate from urban areas. These phenomena indicated urban decline and insufficiencies in resource allocation to meet human needs [64, 65]. These factors contribute to a reduction in greenness in urban areas. Model 5 examined the influence of economic vitality on greenness, indicating that newly established enterprises negatively impact urban greenness (coef = -0.0342, p-value <0.05). Model 5 reveals a significant impact of workers' income on urban greenness. As income levels rise, individuals tend to enhance their living conditions and exhibit a heightened concern for environmental issues [66-68]. This heightened environmental consciousness manifests through activities such as tree planting, environmental improvement around residences [69] and eco-friendly modes of transportation [70]. An increase in income can signify economic development and concurrently contribute to poverty reduction within the community. The study also examined the relationship between the multidimensional poverty rate and income, showing a negative correlation. Subsequently, Model 6 examined the combined effects of population and economic vitality on urban greenness. However, the results only demonstrated the significant impact of

population density and out-migration rates emerging as the primary contributing factors. Furthermore, the study uncovered variations in the effects on urban greenness across different provinces, highlighting the more pronounced reduction in urban greenness within centrally controlled cities such as Ha Noi and Hai Phong. This result aligns with the urban context of Hanoi and Hai Phong. Hanoi and Hai Phong are two major economic centres of the Red River Delta. Trees are decreasing in these cities due to rapid urbanization. The government has seen them as critical economic areas and has established large industrial zones clustered in Hanoi and Hai Phong [71]. High population density can also pressure land funds for residential and commercial purposes, leaving limited space for parks and green areas. The conversion of green areas for parking or new construction projects can contribute to the decline of urban green spaces [72]. This finding aligns with the research conducted by Sun et al [73], who emphasized the close relationship between changes in NDVI and the dynamics of urbanization.

This study investigated the impact of population and economic vitality on rural greenness using models 7, 8, and 9 (Table 4). The findings revealed that the effects of population (model 7) and economic (model 8) factors on greenness were consistent with the observations in models 1 and 2. However, model 9 indicated that rural greenness was influenced by natural, population, and economic factors, whereas natural elements and population characteristics primarily influenced urban greenness. Interestingly, the natural growth rate of the population has a positive impact on rural greenness, which contrasts with the findings of Aburas et al [74], who highlighted a decline in dense vegetation areas due to population growth. In rural areas, agricultural practices (e.g. farming and animal husbandry) enhanced greenness, aligning with the studies conducted by Xu et al [75], Piao et al [76], and Lenney et al [77]. Additionally, non-farm businesses significantly affected rural greenness. These businesses promoted local economic development, created employment opportunities, attracted investment, and reduced poverty, thereby contributing to sustainable economic

development in rural areas [78, 79]. Non-farm businesses encouraged the transition from traditional agricultural production to new business sectors and resource services, alleviating the pressure on natural resources and contributing to environmental protection. However, it should be noted that increased income could lead to a decline in greenness in rural areas. In the Red River Delta, households tend to engage in urbanization activities as they experience economic development and an increase in average income [22]. Consequently, this results in new buildings and infrastructure construction, reducing the agricultural land [80].

Contrary to the conventional expectation that regions with elevated temperature and humidity would exhibit increased greenery, surprisingly, biophysical factors such as temperature and precipitation demonstrate noteworthy negative impacts on urban and rural green spaces. This unexpected finding aligns with the findings of studies conducted by Wanget al [21] and Zhang et al [81], who identified a widespread incongruence between greenness and temperature. The negative influence of these biophysical factors on green spaces may be attributed to the prominence of human activities in driving changes in greenery within developed areas [21]. Broadly, human activities have expedited the dissonance between temperature and greenness [82, 83]. Moreover, climate warming could intensify plants' resource requirements, such as nitrogen, potentially limiting the positive impact of higher temperatures on vegetation growth. Consequently, regions characterized by elevated temperature and precipitation may paradoxically exhibit diminished greenness due to extensive construction activities.

Table 4. The impacts of population and economic vitality on urban and rural greenness

Variables	Urban and rural areas			Urban areas			Rural areas		
	Population vitality (Model 1)	Economic vitality (Model 2)	Population and economic vitality (Model 3)	Population vitality (Model 4)	Economic vitality (Model 5)	Population and economic vitality (Model 6)	Population vitality (Model 7)	Economic vitality (Model 8)	Population and economic vitality (Model 9)
Intercept	1.7229* (0.7422)	4.1633*** (0.7313)	2.228** (0.7755)	2.2170** (0.7726)	4.3747*** (0.7589)	2.5726** (0.8406)	2.1314** (0.8033)	4.9880*** (0.8027)	2.5833** (0.8292)
Population density	-0.0490*** (0.0111)		-0.0632*** (0.0127)	-0.0478*** (0.0116)		-0.0501*** (0.0137)	-0.055*** (0.0121)		-0.0762*** (0.0136)
The natural growth rate of population	-0.0180 (0.0139)		0.0013 (0.0162)	-0.0082 (0.0145)		0.0112 (0.0176)	-0.0091 (0.0151)		0.0348* (0.0174)
In migration rate	-0.0041 (0.0067)		-0.0001 (0.0069)	0.0002 (0.0070)		-0.0026 (0.0075)	-0.0044 (0.0072)		-0.0029 (0.0074)
Out migration rate	0.0350** (0.0107)		0.0267* (0.0105)	-0.0299** (0.0111)		-0.0234* (0.0114)	0.0301* (0.0115)		0.0240* (0.0113)
Number of enterprises		0.0339* (0.0159)	0.0294 # (0.0149)		-0.0342* (0.0165)	-0.0293 # (0.0162)		0.0077 (0.0174)	0.0030 (0.0159)
Foreign investment capital		-0.0127** (0.0045)	-0.0049 (0.0046)		-0.0036 (0.0047)	0.0028 (0.0050)		-0.0104* (0.0050)	-0.0019 (0.0049)
Gross retail sales of goods and services		0.0077 (0.0123)	-0.0205 (0.0128)		0.0135 (0.0128)	-0.0088 (0.0138)		0.0129 (0.0135)	-0.0236 # (0.0136)
Non-farm business		0.0087 (0.0180)	0.0543** (0.0200)		0.0050 (0.0155)	0.0317 (0.0216)		0.0253 (0.0197)	0.0909*** (0.0213)
Workers' income		0.0406** (0.0149)	-0.0367* (0.0154)		0.0406** (0.0149)	0.0009 (0.0167)		-0.0330* (0.0164)	-0.0330* (0.0165)
Temperature	0.1725 (0.216)	-0.7747*** (0.2117)	-0.1879 (0.2412)	0.2115 (0.2251)	-0.4945* (0.2198)	0.0814 (0.2614)	0.0097 (0.2340)	-1.0391*** (0.2198)	-0.3593 (0.2579)
Precipitation	-0.1795*** (0.0400)	-0.1439*** (0.0417)	-0.1271** (0.0413)	-0.2606*** (0.0416)	-0.2654*** (0.0433)	-0.2676*** (0.0448)	-0.1576*** (0.0433)	-0.1550*** (0.0458)	-0.1213** (0.0442)
Centrally controlled cities (1: Yes; 0: No)	-0.014 (0.0136)	-0.064** (0.0193)	-0.0006 (0.002)	-0.1627*** (0.0142)	-0.1196*** (0.0201)	-0.1104*** (0.0234)	-0.0070 (0.0147)	-0.0419 # (0.0212)	0.0287 (-0.0183)
City fixed - effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed - effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	154	154	154	154	154	154	154	154	154
R ²	0.4358	0.3869	0.4939	0.6881	0.663	0.6966	0.3761	0.3027	0.4537

Signif. codes: *** p<0.001; ** p<0.01; * p<0.05; # p<0.1

4. Discussion

Numerous studies have investigated the relationship between population, economy, and greenery [38, 59, 84, 85]. While population growth leads to the conversion of green areas into residential and transportation infrastructure, negatively impacting greenery in urban and rural areas, some studies have found a positive correlation between population density and urban greenery, particularly in cities in the late stages of urbanization or completed the urbanization process [21]. However, in the early and middle stages of urbanization, economic development tends to take precedence over greenness [86, 87]. Vietnam still faces social and environmental challenges due to the unexpected impacts of rapid socio-economic growth [88]. Green spaces have decreased even in small Vietnamese cities [89], indicating inadequate attention to urban greenery. Furthermore, urban green space planning has been hampered by a dearth of foresight and an absence of systematic planning, resulting in an inadequate fulfilment of the demand for green spaces.

This study presents empirical evidence demonstrating a positive correlation between emigration rates and greenness in rural areas, indicating that higher emigration rates benefit greenness. This finding aligns with previous studies conducted by Li et al [90], Preston and Taveras [91], who indicated that the emigration of agricultural labour improves vegetation cover. However, this phenomenon often initiates a migration cycle between regions, as high emigration rates in one area correspond to high immigration rates in another. Excessive immigration can result in densely populated urban areas surpassing their carrying capacity, which is detrimental to urban greenness and may lead to environmental degradation [92]. Conversely, high emigration rates negatively impact urban greenness, contradicting Li et al [90] findings. An excessive emigration rate may indicate either a declining city with insufficient living space, infrastructure, and environmental quality or a dynamic and attractive city [64, 93]. Therefore, achieving a strategic balance between population growth and green spaces is crucial

to promoting sustainable urban development. Further comprehensive studies conducted on a larger scale are needed to understand better the impact of population migration, specifically emigration and immigration rates, on green spaces.

The impact of economic vitality on greenness differs between urban and rural areas in Vietnam. While non-agricultural establishments, labour income, and demographic factors significantly impact rural greenness, demographic factors are the primary influence in urban areas. This finding contradicts the conclusions of Wanget al [21], who found that economic factors, particularly GDP, significantly impact greenness in urban areas. This is because individuals tend to enhance their living conditions and exhibit a heightened concern for environmental issues when the economy is improved [66-68]. In low-developed city centers, the positive impact of the economy on greenness needs to be clarified, which supports the findings of Al-mulali et al [94]. While Al-Mulali et al [95] argued that there is no environmental Kuznets curve since the relationship between the economy and pollution was positive between 1981-2011, this study found that the relationship between the economy and greenness has gradually improved since 2009, particularly in large urban areas like Hanoi and Haiphong. This finding is consistent with Wu and Kim [85], who pointed out that who pointed out that different GS characteristics can characterize different socioeconomic contexts. In the final stage of urbanization, the relationship between the economy and pollution will gradually become negative, indicating a positive relationship between the economy and greenness. This study reveals that economic globalization does not completely lead to environmental degradation, which is consistent with the findings of previous research by Suki et al [96]. It suggests that a threshold exists beyond which the impact of economic development on green space and the environment shifts from negative to positive, as highlighted in the study conducted by Xu et al [97]. Therefore, comprehending the mechanism of the nonlinear relationship between economic development and green space is crucial for a comprehensive understanding of its

effects. Moreover, it is imperative for developing countries to integrate environmental and green space planning into land use planning across all levels in order to foster a mutually beneficial relationship between the economy and greenness. Furthermore, there should be a strong emphasis on environmental monitoring activities, as they play a crucial role in enhancing the environment and facilitating the development of sustainable cities [98]. This integrated approach promotes green cities [99] and facilitates urban-rural economic growth [20].

While economic advancement is undeniably crucial, preserving urban and rural green spaces is equally paramount. The trade-off between immediate economic benefits and the absence of green spaces can have profound implications for urban landscapes and the overall well-being of the populace. The discernible disparity in greenness between central cities and their counterparts underscores the critical importance of proactive urban planning to ensure an adequate provision of green spaces [100]. Urban and rural green area preservation strategies should be implemented in the early land use planning stage. Failure to incorporate such considerations in the early stages of urban development may restrict subsequent opportunities for integrating significant green spaces, necessitating multiple development cycles for substantive greening to manifest.

Various solutions, including preserving green spaces in compact cities and adopting natural-based approaches, merit consideration in rapidly developing urban environments such as Vietnam, facilitating the transition to sustainable cities [101]. Green spaces should be incorporated into redevelopment plans, with specific green space plans formulated prior to the commencement of construction activities [102]. The handling of green spaces during the construction phase is equally crucial, with efforts focused on preserving existing trees or, when unavoidable, transplanting them to suitable locations [103]. Post-redevelopment enhancements to green spaces are essential to optimize functionality, including aesthetic considerations.

Establishing legal frameworks for urban green space planning is indispensable for achieving this objective, necessitating strategic and holistic plans encompassing entire regions, especially in the case of big cities such as Hanoi and Hai Phong.

In Vietnam's rapid urbanization context, effective overall management of green spaces, encompassing both managed and unmanaged vegetation, is necessary for a comprehensive approach that harmonizes greening, conservation, and various land use functions. This approach is vital for addressing historical critiques regarding the loss of natural areas resulting from extensive urban development. Urban planners must pay special attention to the arrangement of green spaces, including parks, streets, and vertical green surfaces, to optimize urban greenery [104]. When ground space is limited, alternative greening strategies present viable options for enhancing greenery in compact cities [100, 105]. Developing green technology solutions, including optimizing resource use and minimizing negative environmental impacts through initiatives such as green roofs, is also deemed crucial [106, 107]. Furthermore, urban planners should incorporate smart city strategies to promote green and low-carbon urban development [108].

5. Conclusion

Green space is an essential component of both urban and rural environments. Through an in-depth exploration of the Red River Delta from 2009 to 2022, the study uncovers how green space changes over time and how population and economic vitality impact greenness. Green space decreased in urban and rural areas from 2009 to 2022, with a notable decline in the initial eight years, followed by a slight recovery. The impact of population and economic vitality on green space differs between urban and rural areas. While the economy and population aspects are the main factors affecting rural green spaces, population characteristics are the main factors affecting green space in urban areas. The results show that the immigration rate and population density are the two main factors influencing greenness in urban and rural

areas. This article reveals that the positive relationship between economic development and greenness in the Red River Delta is becoming more evident. However, this trend is juxtaposed with the negative impact of economic development on greenness in rural areas. Furthermore, significant variations in the influence of population, economic vitality, and greenness are discerned based on the city functions. This study significantly contributes to bridging the gap in international research on green spaces, particularly in the Vietnamese context.

This research is limited by using NDVI to represent green space in urban and rural areas. NDVI may not differentiate between different types of green space and the ecological characteristics of green space. Furthermore, this study only focuses on the one-way relationship between population, economic vitality, and green space in urban and rural areas. Future research could explore the two-way relationship between them. Additionally, this study does not focus on spatial heterogeneity between provinces. Future works can use Geographically Weighted Regression to explore the spatial effects of population and economic dimensions on greenness. Despite these limitations, the study contributes to a broader understanding of urban growth theory, focusing on green space dynamics based on city type and function. It also offers valuable insights into developing cities in the early stages of urbanization.

6. References

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